

WASP-52b

R-0298

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_231205 · created 2026-07-17T20:03:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460283.6196 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.015
Transit depth (Rp/Rs)^2	2.76 +/- 0.51 [%]
Orbital Inclination (inc)	86.16 +/- 1.59
Airmass coefficient 1 (a1)	0.932 +/- 0.014
Airmass coefficient 2 (a2)	0.052 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	#1 - [420, 178]
Optimal Aperture	2.63
Optimal Annulus	9.42
Transit Duration (day)	0.0782 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.015 vs 0.1646 (deviation 0.09σ)
Duration fit vs published	0.0782 d vs 0.0758 d (deviation 0.29σ)
Mid-time O–C	-0.5 min
Depth SNR	11.1
Residual scatter	1.52 %

Light curve

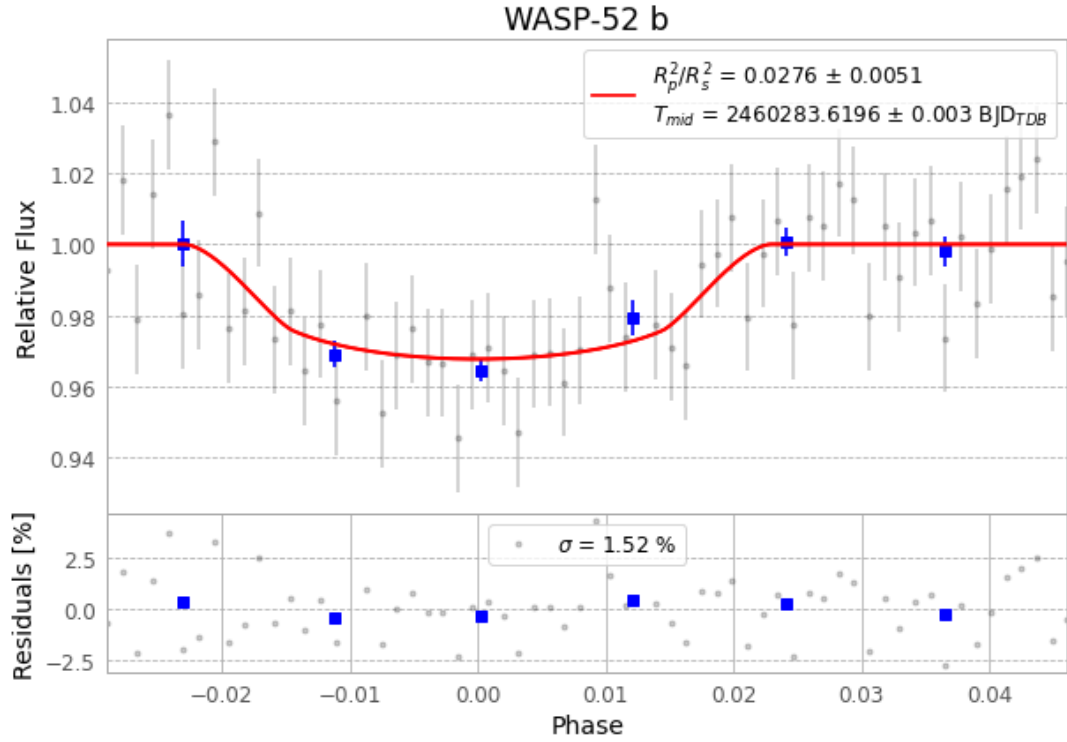


Figure 1 — FinalLightCurve_WASP-52 b_2023-12-04.png (SHA-256 a7f0ce77004b892d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [310, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8ad6ccf6dc4eb2e9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3612b4b

Data provenance

64 FITS frames (42 MB), WASP-52231205013615.FITS → WASP-52231205044515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-231205.log (2235a9dcc7d8...), FinalLightCurve_WASP-52 b_2023-12-04.png (a7f0ce77004b...), FinalLightCurve_WASP-52 b_2023-12-04.csv (c0692c2e029c...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 20:03Z.